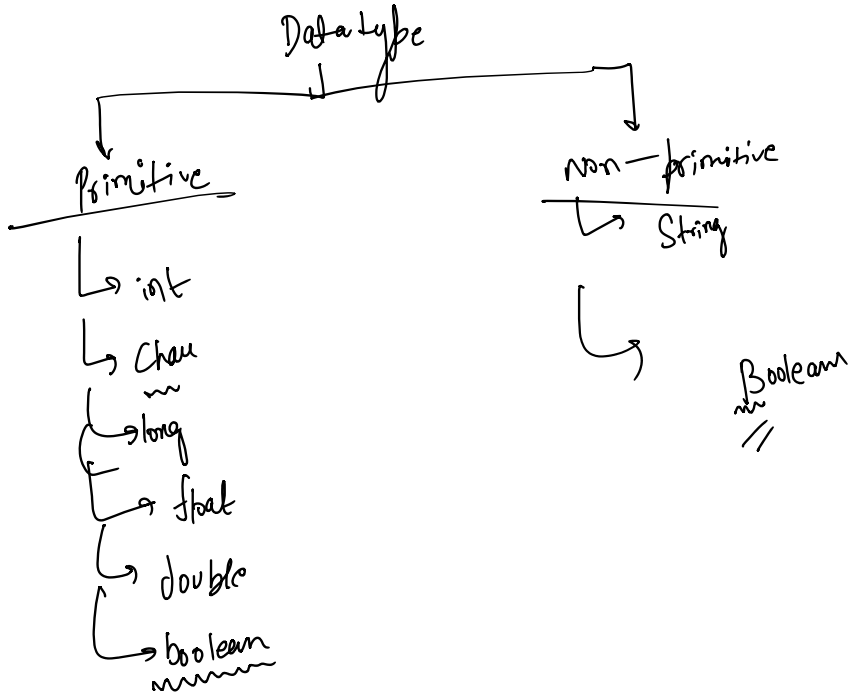


variables = ? $\rightarrow$ Container



- Rules to define a variable.

① Don't start your variable with a number

int num ✓
int 1num ✗

② Don't use keywords.

`public static void main() {`

③ variables are case-sensitive.

int num; { static (keyword)
int Num; { int Static ✓

int Num; int State (x)

④ Use special characters or numbers in between
(not in the first place);

class Maths = 2

type conversion

→ implicit
→ explicit

int → 4 byte
short → 2 byte ✓
byte → 1 byte
long → 8 byte

1 byte = 8 bits

int = 8 × 4 = 32

-2^{32-1} to $2^{32-1} - 1$

↳ -2^{31} to $2^{31} - 1$

2 byte
↳ 16

short

-2^{15} to $2^{15} - 1$

Byte

-128 to 127

$2^7 = 128$

↳ -2^{31} to $2^{31} - 1$
↳ -128 to 127

byte b = 36;

int a = b;

cout << a;

op → 36

1) int a = 128;

int a = 32;

byte b = (byte) a;

int a = ...

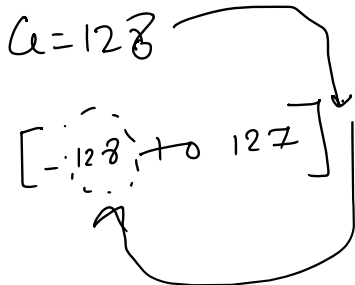
byte b = (byte) a;

→ explicit type conversion

byte b = (byte) a;

cout(b);

$$128 = 127 + 1 \rightarrow 129$$



-128, -127, -126, ..., 0, 1, 2, ..., 127

int a = 300;

byte b = (byte) a;

cout(b);

$$300 \% 256$$

$$= 44$$

Q. byte a = 2;

int b = 27;

cout(a+b);

→ 29
→ (int)

byte -
Short -
int

→ byte a = 2;

→ byte b = 2

X byte c = a * b;

→ byte c = (byte) (a * b);

byte a = 100

byte b = 100

byte c = a * b

$$2 \times 2 = 4$$

byte -

* Imp points

- byte, short, char values are promoted to int
- If one operand is long, then whole expression will become long.
- If one operand float, then whole expression float
- If one operand double whole expression double.

float = 4 byte

double = 8 byte

float f = 3.14 f

double d = 3.14 d;

char ch = 'a';
 ↳ ASCII
 ↳ 97 → a;

Q. cout (7+2+'a'); → char

cout (7+2+"H");
 ↳ String

↳ float a = 3/1.0 f;

double a = 3/1.0;

f
 d
 1.0
 ↳ double

Q. cout ((double)10/3);

Q. add two binary numbers and print the result (in decimal).

↳ 10001

1 0 0 1

→ 10001

int bin = 10001

int ans = 0; temp = 0;

while (bin > 0) {

int multiplier = bin % 10;

ans = ans + multiplier * Math.pow(2, temp);

temp++;

→ bin = bin / 10;

}

1 0 0 1

3 2 1 0

1×2^0

$+ 0 \times 2^1$

$+ 0 \times 2^2$

$+ 1 \times 2^3$

pow(base, power)

$(base)^{power}$